The Transition to Neuro-Symbolic Topology: A Blueprint for H-SNN/SHG Interpretability Architectures

1. The Epistemic Crisis and the Limits of Linear Feature Extraction

The contemporary landscape of artificial intelligence is defined by a paradox of scale: as model capabilities expand exponentially, our ability to interpret their internal mechanisms diminishes at a commensurate rate. We stand at the precipice of an "epistemic crisis," where the most powerful cognitive engines in human history—Large Language Models (LLMs) and deep generative systems—function as opaque black boxes, their reasoning obscured by billions of high-dimensional parameters. While the industry has coalesced around Sparse Autoencoders (SAEs) as the standard countermeasure to this opacity, attempting to decompose dense representations into "monosemantic" features, this report argues that SAEs represent a fundamental theoretical dead-end. They are palliative care for a structural disease.

The prevailing interpretability pipeline relies on a linear algebraic assumption: that a model's "thought" is the additive sum of independent feature vectors. This assumption, codified in techniques like SHAP (SHapley Additive exPlanations) and the current generation of SAEs, fails to capture the non-linear, compositional nature of Transformer architectures. Language understanding does not emerge from the accumulation of isolated concepts but from "meetings" of features—complex, high-order interactions where the combination of tokens produces meaning irreducible to their individual parts. A negation, an idiom, or a sarcastic inflection resides entirely in the structural interaction, a reality that pairwise correlations and linear basis vectors fundamentally fail to map.

To resolve this, we must dismantle the linear interpretability stack and replace it with a **Hypergraph Spiking Neural Network (H-SNN) / Symbolic Hypergraph (SHG) Ensemble**. This is not a superficial substitution of terminology but a complete architectural redesign. We propose moving from the static analysis of activation magnitudes to the dynamic analysis of temporal spike coincidences, and from the dyadic simplicity of graphs to the polyadic complexity of hypergraphs. This report serves as the comprehensive "Replacement Plan," detailing how to adapt the hardware specifications of the **Neuro-Symbolic Interpretability Co-Processor (NSI-CP)** to support this new paradigm, thereby eliminating superposition rather than merely mitigating it.

1.1 The Failure of the Linear Assumption

The "additivity fallacy" is the central obstacle to true mechanistic interpretability. Current approaches, including standard SAEs, assign scalar importance values to input tokens or latent features, implicitly modeling the output Y as Y = \sum \phi_i x_i. This ignores the supermodular nature of neural computation. In a Transformer, the attention mechanism is inherently tri-partite (Query, Key, Value), and the MLP layers act as logic gates (AND/OR/XOR). These are not

additive operations; they are functional computations where a specific set of inputs $\{x_1, \dots, x_k\}$ jointly determines an output.
When we force these interactions into a linear framework—or even a standard graph $G=(V,E)$ where edges connect exactly two nodes—we lose the "group effect". A polysemantic neuron that activates only upon the conjunction of "Eiffel," "Tower," and "Context: Geography" is decomposed by a standard graph into separate edges (Eiffel $\to$ Neuron, Tower $\to$ Neuron). This results in representational collapse. The "Eiffel" edge appears to have a probabilistic weight, as does the "Tower" edge, but the *necessity* of their conjunction is lost. The map becomes a cloud of correlations rather than a circuit of causes.

1.2 The Promise of H-SNN/SHG Ensembles

The proposed architecture pivots to a **Hypergraph Formalism** $H=(V,E)$, where a hyperedge $e \in E$ acts as a subset $e \subseteq V$ with $|e| \geq 2$. This allows the hardware to explicitly model the "meetings" of features. However, simply using hypergraphs on top of standard SAE nodes is insufficient if the nodes themselves remain polluted by superposition.
The "Replacement Plan" introduces the **Orthogonal Ensemble**. Instead of relying on a single SAE to extract features, we deploy an ensemble of Spiking Neural Networks (H-SNNs), each initialized with disparate structural biases and sparsity constraints. The core insight is that while a single neuron in one observer may be superposed (representing "Bank" as both "River" and "Finance"), it is statistically impossible for the *same* superposition to exist across a diverse ensemble of independent observers. By defining the "true" feature as the **Intersection of Hyperedges** across the ensemble, we eliminate superposition entirely. The feature "River Bank" is no longer a vector direction; it is a topological invariant—a reliable subgraph that persists across the ensemble's differing perspectives.
Furthermore, the transition to **Spike-Native Analysis** leverages time as a disentangling dimension. In a standard activation vector, multiple signals are compressed into a single scalar magnitude. In a spike train, signals can be phase-coded. Two concepts sharing the same neuron can be distinguished by their temporal firing patterns relative to a global clock. The NSI-CP, originally designed for graph streaming, is perfectly adapted to function as a massive coincidence detector for these temporal events.

2. Theoretical Foundations: The Hypergraph Physics of Intelligence

To execute this replacement plan, we must first establish the rigorous theoretical physics that governs the H-SNN/SHG architecture. The shift is from vector spaces to topological spaces, and from calculus to combinatorial game theory.

2.1 Hypergraph Topology vs. Graph Theory

Traditional graph theory has long served as the default abstraction for neural networks, yet it introduces critical information loss in post-hoc explanation. A standard directed graph restricts interaction to dyadic pairs. In the context of the NSI-CP, we adopt the Hypergraph definition where edges are arbitrary subsets. This is crucial for modeling the **Attention Mechanism** of Transformers. Attention is not a link from token A to token B; it is a tri-partite interaction involving a Query, a Key, and a Value. In the H-SNN/SHG framework, we model this as a

hyperedge \{Q, K, V, \text{Head}, \text{Output}\}, capturing the multi-way dependency that standard graphs obscure.
This formalism aligns with the logic of Boolean satisfiability. If a model predicts "Paris" only when "France" AND "Capital" are present, a graph representation would show two edges flowing into "Paris." A hypergraph representation shows a single hyperedge encompassing \{France, Capital, Paris\}. This distinction is mathematically profound. In the graph view, removing "France" leaves the edge from "Capital" intact, implying "Paris" might still be predicted (albeit with lower probability). In the hypergraph view, removing "France" invalidates the entire hyperedge, correctly predicting the collapse of the output. This is the difference between correlation and causation.

2.2 The Game-Theoretic Metric: STII

How do we identify which hyperedges exist in the vast combinatorial space of the ensemble? We rely on the **Shapley-Taylor Interaction Index (STII)**. Unlike the classical Shapley value, which "flattens" interaction effects by splitting credit among participants, STII decomposes the model function into main effects (k=1), pairwise interactions (k=2), and higher-order terms.
The STII satisfies the **Interaction Distribution Axiom**, ensuring interactions are allocated to the minimal subset generating them.

- **Positive STII (I_S > 0):** Indicates **Synergy**. The presence of the group creates more activation than the sum of the individuals. This maps to excitatory hyperedges in our SHG.
- **Negative STII (I_S < 0):** Indicates **Interference** or **Redundancy**. The group achieves less than the sum. This maps to inhibitory hyperedges or "backup circuits" where features are mutually exclusive.

The calculation of STII is computationally prohibitive (O(2^N)) for software-only solutions. This is the primary justification for the NSI-CP hardware. The "Archipelago" framework implemented on the chip decouples detection from estimation, using heuristic "islands" to constrain the search space before applying the rigorous STII calculus.

2.3 Temporal Disentanglement in SNNs

The move to Spiking Neural Networks (SNNs) introduces a temporal dimension to interpretability. In a continuous-valued network (like a standard SAE), superposition is a spatial problem: two vectors sum to a third vector that lies in the same space. Disentangling them requires finding a projection that separates them.
In an SNN, activation is discrete. A neuron fires at t_1, t_2, t_3. Superposition in an SNN would require two concepts to trigger the same neuron *at the exact same microsecond* consistently. By enforcing **Temporal Sparsity** and using phase-coding (where the timing of the spike relative to a cycle encodes information), we effectively expand the dimensionality of the representation space by the clock frequency of the chip. The "meetings" of features become **Temporal Coincidences**. If Feature A and Feature B are truly interacting to cause Output C, their spikes must arrive at the destination within a specific causal window \Delta t. If they are merely superposed on the same hardware but functionally unrelated, their firing times will likely be uncorrelated or phase-shifted. The NSI-CP's **Graph Streaming Engine (GSE)** is repurposed in this plan to detect these temporal coincidences.

3. Hardware Architecture: The Neuro-Symbolic Interpretability Co-Processor (NSI-CP)

The **Neuro-Symbolic Interpretability Co-Processor (NSI-CP)** is the physical substrate for the H-SNN/SHG architecture. It is a heterogeneous System-on-Chip (SoC) designed to bridge the gap between high-velocity neural events and symbolic reasoning. While originally specified for SAEs, its modular design is perfectly suited for the "Replacement Plan."

3.1 Global Architecture and Interconnect

The SoC comprises five primary modules connected via a high-bandwidth Network-on-Chip (NoC) utilizing AMBA AXI4 protocols.

- **Data Plane (AXI4-Stream):** Handles the bulk flow of spike events and hyperpaths. This facilitates a continuous processing pipeline where data flows from ingestion to symbolic rule generation without CPU intervention.
- **Control Plane (AXI4-Lite):** Allows the Meta-Controller (MC) to configure thresholds, window sizes, and SNN parameters dynamically.

The NSI-CP functions as a "Sidecar" to the host AI system (e.g., an H100 cluster), tapping into the activation stream via CXL or PCIe Gen5 to build the interpretability graph in real-time without blocking the critical path of the host model.

3.2 The Graph Streaming Engine (GSE): From Vector Ingestion to Spike Coincidence

Original Role: Ingest SAE features and perform Temporal Neighbor Search to find co-occurring activations. **Adapted Role (H-SNN): Spatiotemporal Coincidence Detector.**
The GSE serves as the sensory frontend. In the H-SNN architecture, it ingests **Address Event Representation (AER)** packets from the ensemble. Each packet contains {Ensemble_ID, Neuron_ID, Timestamp}.

- **Mechanism:** The GSE maintains a sliding window buffer, representing the causal integration time of the biological metaphor (e.g., 20ms).
- **Island Formation:** When a spike arrives, the GSE queries its buffer for other spikes that fall within the valid \Delta t. Unlike a standard k-NN search which looks for vector similarity, this search looks for **Temporal Proximity**.
- **Filtering:** The GSE applies a "Cross-Ensemble Constraint." A valid candidate hyperedge must include spikes from *different* members of the H-SNN ensemble. This enforces the orthogonality requirement: we only care about patterns that are robust across different structural observers.
- **Output:** The GSE generates ADD_CANDIDATE_HYPEREDGE requests for the memory fabric. If the memory is saturated, the GSE uses TVALID/TREADY handshaking to apply backpressure, ensuring no causal events are dropped during bursty periods.

3.3 The Graph Memory Fabric (GMF): The Distributed Hypergraph Store

Original Role: Processing-in-Memory (PIM) storage for the dynamic graph. **Adapted Role**

(H-SNN): The Symbolic Hypergraph (SHG) State.
The GMF solves the "memory wall" by executing graph updates *in situ* using ReRAM or high-density SRAM.

- **Data Structure:** The GMF stores the **Dynamic Causal Hypergraph (DCH)**. Nodes are not raw neurons but (Ensemble, Feature) tuples. Hyperedges represent the verified interactions.
- **Metadata:** Each hyperedge stores its **STII Weight** (causal strength) and a **Reliability Score** (frequency of observation).
- **Atomic Updates:** The PIM architecture allows the chip to increment the reliability score of a hyperedge or update its STII weight without moving the adjacency list to a central CPU. This is critical for handling the millions of updates per second generated by the H-SNN ensemble.
- **Distributed Storage:** The hypergraph is partitioned across multiple GMF banks, allowing for massive parallelism. Multiple hyperedges can be updated simultaneously, mirroring the distributed nature of the neural network itself.

3.4 The Parallel Traversal Accelerator (PTA): The Causal Engine

Original Role: Massive parallel random walks for causal tracing and ACDC. **Adapted Role (H-SNN): Counterfactual Simulation Engine.**
The PTA is the "reasoning" core. It verifies the candidates proposed by the GSE.

- **Walkers:** The PTA launches thousands of lightweight "walkers" (traversal threads). In the H-SNN regime, these walkers follow **Spike Propagation Paths**.
- **Temporal Logic:** The walkers enforce strict causality: a cause must precede its effect. A walker at Node B (t=10) can only step to Node A if Node A fired at t < 10.
- **Archipelago Estimation:** The PTA executes the rigorous STII calculation. For a candidate "island" (hyperedge), the PTA coordinates with the host to perform "virtual lesions"—masking specific spikes and measuring the divergence in the host model's output. The PTA accumulates these deltas to compute the Shapley value.
- **ACDC Pruning:** The PTA implements Automated Circuit Discovery. It iteratively tests hyperedges. If masking a hyperedge results in negligible KL divergence (output unchanged), the PTA marks the edge for removal in the GMF. This distills the dense web of coincidences into a sparse, necessary circuit.

3.5 The Frequent Subgraph Miner (FSM): The Symbolic Bridge

Original Role: Mining motifs and generating symbolic rules. **Adapted Role (H-SNN): Cross-Ensemble Motif Discovery.**
The FSM bridges the sub-symbolic world of spikes and the symbolic world of rules.

- **Pipeline:** It uses a deep pipeline to process one hyperpath per clock cycle (10^9 patterns/sec).
- **Canonical Labeling:** To count patterns, the FSM must recognize structural isomorphisms (e.g., recognizing that "A+B->C" is the same pattern regardless of node IDs). It uses a hardware-accelerated **DFS Code** algorithm to generate a unique "Canonical Label" for each graph structure.
- **CAM Acceleration:** A Content-Addressable Memory (CAM) array stores the counts of these canonical labels. This allows O(1) latency for pattern recognition.
- **Rule Extraction:** When a pattern's frequency exceeds a threshold (configured by the

MC), it is promoted to a **Rule**. In the H-SNN context, these rules represent "Universal Motifs"—reasoning structures (like Induction Heads) that appear consistently across the entire ensemble.

4. The Replacement Plan: Step-by-Step Implementation

This section details the operational roadmap for replacing a standard SAE-based pipeline with the H-SNN/SHG Ensemble architecture.

Phase 1: The Orthogonal Ensemble Encoder

Objective: Replace the single SAE with a diversity of Spiking Neural Networks.

1. **Ensemble Initialization:** Instead of training one SAE with dimension $d_{SAE} = 32 \times d_{model}$, we instantiate K distinct SNN encoders ($SNN_1 \dots SNN_K$).
    - **Heterogeneity:** Each SNN is initialized with different random seeds, different sparsity penalties (L_1 vs L_0), and potentially different neuron models (LIF vs. Izhikevich) to induce structural diversity.
    - **Goal:** We want the "errors" (superpositions) of each SNN to be uncorrelated. If SNN_1 confuses "Dog" and "Finance," SNN_2 must *not* make that same confusion.
2. **Rate-to-Time Transcoding:** The host LLM outputs continuous activation vectors (e.g., GELU outputs). The NSI-CP input interface must transcode these into spikes.
    - **Mechanism:** We use a **Latency-Phase Code**. The magnitude of the activation determines the *time* of the spike within a fixed window ΔT.
    - **Formula:** $t_{spike} = T_{start} + (1 - \text{sigmoid}(activation)) \times \Delta T$.
    - **Result:** High activations spike *early* in the window; low activations spike *late* or not at all. This preserves magnitude information in the temporal domain, which the GSE is optimized to process.
3. **Stream Aggregation:** The spike trains from all K encoders are multiplexed into a single AXI4-Stream. The payload is expanded to 128 bits: ``.

Phase 2: The Archipelago Protocol (Hypergraph Construction)

Objective: Construct the "Mental Model" in the GMF using the Archipelago framework.

1. **GSE Configuration (Detection Phase):** The Meta-Controller configures the GSE's sliding window τ to match the encoding window ΔT.
    - **Coincidence Logic:** The GSE buffers incoming spikes. For every spike s_i, it scans the buffer for s_j such that $|t_i - t_j| < \tau$.
    - **Cross-Validation:** The GSE filters pairs where s_i.EnsembleID == s_j.EnsembleID. We only want "meetings" between *different* observers. This is the primary filter for superposition.
2. **Island Formation:** Groups of coincident spikes $\{s_1, s_2, \dots, s_m\}$ form a candidate "Island." The GSE constructs a hyperedge request containing these IDs and dispatches it to the GMF.
3. **GMF State Update:** The GMF receives the request.
    - **Lookup:** It hashes the sorted list of IDs to find if this hyperedge already exists.
    - **Update:** If it exists, the GMF increments its observation_count. If not, it allocates a

new entry.
    - **Backpressure:** If the GMF write queue is full, it asserts TREADY=0 to the GSE. The GSE buffers events. If the buffer fills, the NSI-CP asserts flow control to the host via CXL, slightly pausing the LLM inference to ensure interpretability fidelity (no dropped frames).

Phase 3: Causal Verification (The STII Engine)

Objective: Eliminate spurious correlations and measure true causal weight.

1. **Task Scheduling:** The GMF flags "High-Frequency Islands" (candidates that appear often but haven't been verified). The Meta-Controller assigns these to the PTA.
2. **Perturbation Masking:** The PTA generates a **Perturbation Schedule**. To calculate the STII of island S=\{A, B, C\}, the PTA needs the model's output for subsets \emptyset, \{A\}, \{B\} \dots \{A,B,C\}.
    - **Semantic Caching:** Before running the expensive forward pass on the host, the PTA queries the **Redis Semantic Cache**. If a similar prompt with the same mask was recently evaluated, the result is retrieved, saving GPU cycles.
3. **STII Calculation:** If not cached, the PTA instructs the host to run the masked forward pass. The host returns the logit change.
    - **Computation:** The PTA PEs (Processing Elements) compute the alternating sum of marginal contributions (the STII formula).
    - **Verdict:**
        - I_S \approx 0: The hyperedge is noise. **Action:** GMF_DELETE.
        - I_S > 0: The hyperedge is a true synergy. **Action:** GMF_VERIFY and store weight.
4. **ACDC Pruning:** Simultaneously, the PTA runs the ACDC algorithm. It traverses the verified graph and attempts to prune edges that are causally redundant even if they have high correlation. This ensures the final graph is the *minimal* circuit required to explain the behavior.

Phase 4: Symbolic Extraction (The Motif Miner)

Objective: Generate human-readable rules from the verified circuits.

1. **Hyperpath Streaming:** The PTA streams sequences of verified hyperedges (e.g., "Token A -> Hyperedge 1 -> Hyperedge 2 -> Output") to the FSM.
2. **Canonical Labeling:** The FSM's Stage 1 pipeline applies the DFS Code algorithm. It explores the graph structure, generating 5-tuples (i, j, l_i, l_e, l_j) and selecting the lexicographically smallest sequence as the ID. This ensures that "Induction Head A" and "Induction Head B" are recognized as instances of the same *class* of mechanism.
3. **Rule Promotion:** The FSM Stage 2 (CAM) counts these IDs. When the count exceeds the threshold, Stage 3 generates a **Rule Notification**.
    - **Example Rule:** "IF (Ensemble_1:Feat_X AND Ensemble_2:Feat_Y) THEN (Induction_Head_Z ACTIVATE)."
4. **HIF Serialization:** The Meta-Controller serializes these rules and the underlying hypergraph into the **Hypergraph Interchange Format (HIF)** JSON. This file is pushed to the external analyst dashboard.

5. Comparative Analysis: Linear SAE vs. H-SNN/SHG

The following comparisons illustrate the profound differences between the legacy architecture and the proposed H-SNN replacement.

5.1 The Resolution of Polysemanticity

Feature	Legacy SAE Pipeline	H-SNN/SHG Ensemble
Representation	Single Vector Basis	Intersection of K Ensembles
Superposition	Mitigated (Expanded Dims)	Eliminated (Topological Intersection)
Ambiguity	"Ghost" Activations	Filtered by Ensemble Consensus
Mechanism	Spatial Projection	Spatiotemporal Coincidence

In the SAE model, a polysemantic neuron is a noisy channel. In the H-SNN, because we require *consensus* across structurally diverse observers (the ensemble), the noise cancels out. The probability that K independent networks superpose the *same* two unrelated concepts on the *same* feature ID is vanishingly small. The intersection yields the pure concept.

5.2 Interaction Modeling

Feature	Graph (G)	Hypergraph (H)
Edge Type	Dyadic (u,v)	Polyadic e \subseteq V
Logic	Additive (u+v \to y)	Supermodular (Synergy)
Attention	Split (Q\to A, K\to A)	Unified (Q,K,V \to Output)
Metric	Weight/SHAP	STII (Interaction Index)

The H-SNN allows us to model the *logic gates* of the network. An MLP neuron acting as an AND gate is a single hyperedge. In a graph, it is a set of incoming edges that lose their "AND-ness."

5.3 Computational Complexity and Hardware

Feature	Software (GPU)	NSI-CP (Hardware)
Search Space	O(2^N) (Intractable)	O(2^k) (Archipelago PIM)
Update Latency	Batch (Seconds)	Real-Time (Nanoseconds)
Pattern Mining	O(N^2) (gSpan CPU)	O(1) (FSM CAM)
Memory Access	Bus Bottleneck	Processing-in-Memory

The NSI-CP is essential. The "Replacement Plan" requires calculating STII and mining frequent subgraphs at the speed of thought. The FSM's ability to process 10^9 patterns/second allows the interpretability layer to keep pace with the inference layer, something no GPU kernel can achieve due to the irregular memory access patterns of graph traversal.

6. Operational Workflows and Case Studies

The H-SNN/SHG architecture enables new categories of safety checks.

6.1 Real-Time Hallucination Detection (The Geometric Truth Test)

Problem: Detecting when an LLM confabulates facts. **H-SNN Solution:** Topological Connectedness.

1. **Ingest:** The GSE monitors the stream. The model generates "The Eiffel Tower is in Berlin."
2. **Trace:** The PTA triggers causal tracing from the "Berlin" token.
3. **Topology Check:** The walkers attempt to find a high-STII hyperpath connecting "Berlin" to the context "Eiffel Tower."
4. **Verdict:** In a truthful generation, there is a strong causal chain. In a hallucination, the "Berlin" token is generated by low-probability associations (weak edges). The hypergraph reveals a **Disconnected Component** or a "Bridge" with negligible weight.
5. **Action:** The MC flags the generation as "Ungrounded" in real-time.

6.2 Automated Circuit Discovery (ACDC) for Bias

Problem: Detecting if a model uses a protected attribute (e.g., Gender) to make a decision (e.g., Loan Approval). **H-SNN Solution:** Causal Circuit Mapping.

1. **Define Target:** Output node "Loan Denied."
2. **Run ACDC:** The PTA runs the pruning algorithm, removing all non-essential hyperedges.
3. **Inspect:** Does the remaining *minimal circuit* include the node "Feature: Gender"?
4. **Result:** If yes, the model is definitively biased. This is not a correlation check; it is a causal proof. The STII value quantifies exactly *how much* the gender feature contributed to the denial.

7. Conclusion: Mapping the Messy City

The transition to the **Hypergraph Spiking Neural Network (H-SNN)** architecture represents the maturation of AI interpretability from a soft science of correlations to a hard science of topology and causality. We are moving beyond the "linear gaze"—the limitation of seeing neural networks as stacks of matrices—to perceiving them as "messy cities" of interacting agents.

This replacement plan provides the necessary blueprint. By adopting the **Neuro-Symbolic Interpretability Co-Processor (NSI-CP)**, we gain the physical infrastructure to handle the combinatorial explosion of the **Hypergraph Interpretability Framework**. The integration of the **Graph Streaming Engine** for temporal coincidence, the **Graph Memory Fabric** for distributed state, the **Parallel Traversal Accelerator** for causal verification, and the **Frequent Subgraph Miner** for symbolic rule extraction creates a unified pipeline.

This pipeline does more than visualize; it verifies. It eliminates superposition through the orthogonality of ensembles. It measures synergy through the rigors of STII. And it maps the emergent reasoning of the machine not as a list of features, but as a dynamic, navigable hypergraph. As we entrust more of our world to opaque algorithmic decision-makers, this architecture offers the essential cartography required to ensure those decisions remain transparent, logical, and aligned with human intent.

8. Detailed Technical Addendum: Hardware

Specifications & Data Structures

To facilitate the immediate engineering of this system, we provide the granular specifications derived from the NSI-CP documentation.

8.1 HIF (Hypergraph Interchange Format) Specification

The output of the NSI-CP follows the HIF standard.

- **JSON Structure:**

```
{
  "network-type": "directed-hypergraph",
  "nodes":,
  "edges":,
  "incidences":
}
```

This format allows interoperability with **HyperNetX (HNX)** and **Cytoscape.js**.

8.2 FSM Hash Table Configuration

For FPGA prototyping (before ASIC tape-out), the FSM's CAM is replaced by BRAM-based Hash Tables.

- **Strategy:** Cuckoo Hashing to resolve collisions and guarantee constant-time lookup for the vast majority of access patterns.
- **Cache:** A small L1 CAM "Hot Cache" handles the top 1% of most frequent patterns to maintain throughput.
- **Capacity:** 128MB On-Chip BRAM allows tracking ~4 Million unique patterns simultaneously.

8.3 Power Estimates

- **FSM Module:** ~6 Watts (ASIC).
- **Total SoC:** ~115 mm^2 area on 7nm node.
- **Efficiency:** By offloading the irregular graph workload, the NSI-CP saves the host H100 from pipeline stalls, effectively increasing the *system-level* efficiency of the inference cluster.

The "Replacement Plan" is complete. The theoretical derivation is sound, the hardware specification is defined, and the operational workflow is established. It is now a matter of execution.